GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

229285.1.pdf *Geochronology Record 1911*

229285.1: actinolite–biotite schist, Mount Rupert

(Southern Cross Supergroup; Youanmi Terrane, Yilgarn Craton)

SS Romano, Y Lu, IOH Fielding, MTD Wingate and TJ Ivanic

Location and sampling

MOORA SH50-10, WONGAN 2236 Zone 50, MGA 465084E 6596128N
Warox Site TJISWT000004 Sampled on 24 August 2020

This sample was collected from the 239.15 – 240.60 m depth interval of diamond drillcore MRRC0024, drilled by Red River Resources Ltd at their Wongan Hills project. The drillhole is located about 11.5 km west of the Northam Pithara Road, 6.6 km south-southwest of Little Wongan Hill, and 1.2 km north of Mount Rupert.

Geological context

The unit sampled is an amphibole–biotite schist from the Wongan Hill greenstone belt, Southern Cross Supergroup in the Youanmi Terrane of the Yilgarn Craton (Quentin de Gromard et al., 2021; Ivanic et al., 2022). The Southern Cross Supergroup consists of metamorphosed basalt, andesite–dacite–rhyolite, and variable proportions of sedimentary rocks exposed in several belts along a northwest-trending, 750 km-long corridor from Ravensthorpe in the south to the Weld Range in the north within the Youanmi Terrane (Ivanic et al., 2022). This supergroup contain six formations, i.e. the Wongan, Madoonga, Gossan Hill, Mount Gibson, Honman, and Annabelle Volcanics Formations, which are unconformably overlain by the c. 2.8 Ga Norie and Polelle Groups of the Murchison Supergroup (Van Kranendonk et al., 2013; Ivanic et al., 2022). This sample yielded a maximum depositional age of 3006 ± 6 Ma (see below), which suggests a stratigraphic position at the bottom of the Southern Cross Supergroup. A metamonzogranitic dyke sampled about 16.9 km to the south yielded an igneous crystallization age of 2670 ± 7 Ma with 3219–2785 Ma for inherited zircon cores (GSWA 205933, Wingate et al 2018). Pidgeon et al. (1990) reported an age of 3010 ± 7 Ma for crystallization of a porphyritic felsic rock within the Wongan Hills belt. These authors also reported an age of 2651 ± 4 Ma for post-tectonic porphyritic granite east of the belt, and age components of c. 3.0 and 2.8 Ga for zircons within granitic gneiss south of the belt.

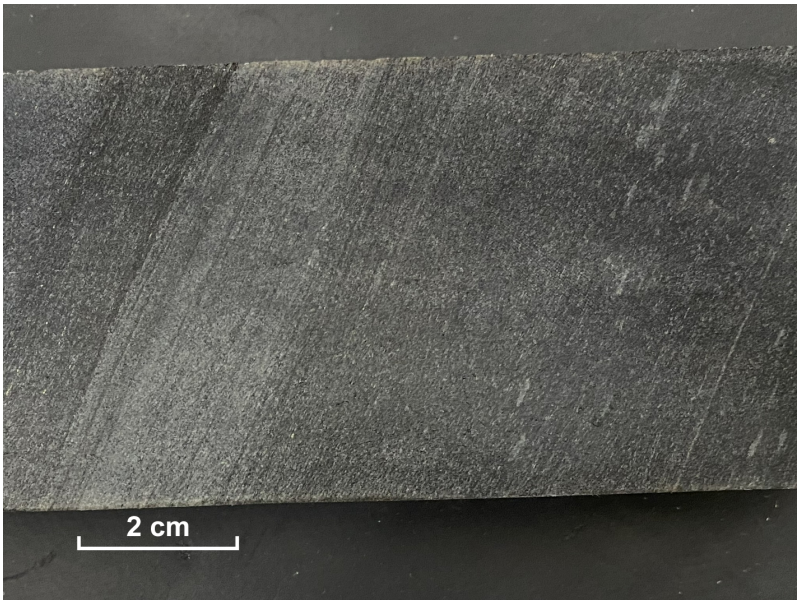

Figure 1. Drillcore image of sample 229285: actinolite–biotite schist, Mount Rupert

Petrographic description

The sample is a foliated and compositionally layered, actinolite–biotite schist, consisting of about 45–50% quartz, 35% actinolite, 15% biotite, 3% magnetite, <1% sulfides and accessory plagioclase, apatite and zircon. Compositional layering is

defined by biotite-rich layers up to 7 mm wide with average quartz grain size of ~0.25 mm, and biotite-poor domains with smaller quartz grains (~0.1 mm). Quartz occurs as xenoblastic and elongate grains up to 0.5 mm in length. Amphibole is colourless to pale green, subhedral, and elongate to 2 mm in length. Subhedral brown biotite (<1 mm) is strongly aligned within the foliation. Magnetite occurs as elongated or equant, subhedral grains to 0.2 mm long. Sulfides, dominated by chalcopyrite and traces of pyrite and pyrrhotite, form clusters (<0.3 mm) or rare lenses (<1 mm) that are aligned parallel to the foliation. Apatite occurs as bladed euhedral laths (0.15 mm). Metamorphic grade is amphibolite facies and the protolith is considered to be a thinly bedded Fe–Mg-rich sedimentary rock.

Zircon morphology

Zircons isolated from this sample are colourless to pale brown, anhedral to subhedral, and variably rounded. The crystals are up to 300 µm long with aspect ratios up to 3:1. In cathodoluminescence (CL) images, concentric or planar zoning is ubiquitous, and few grains show discordant or metamict cores. A CL image of zircons is shown in Figure 2.

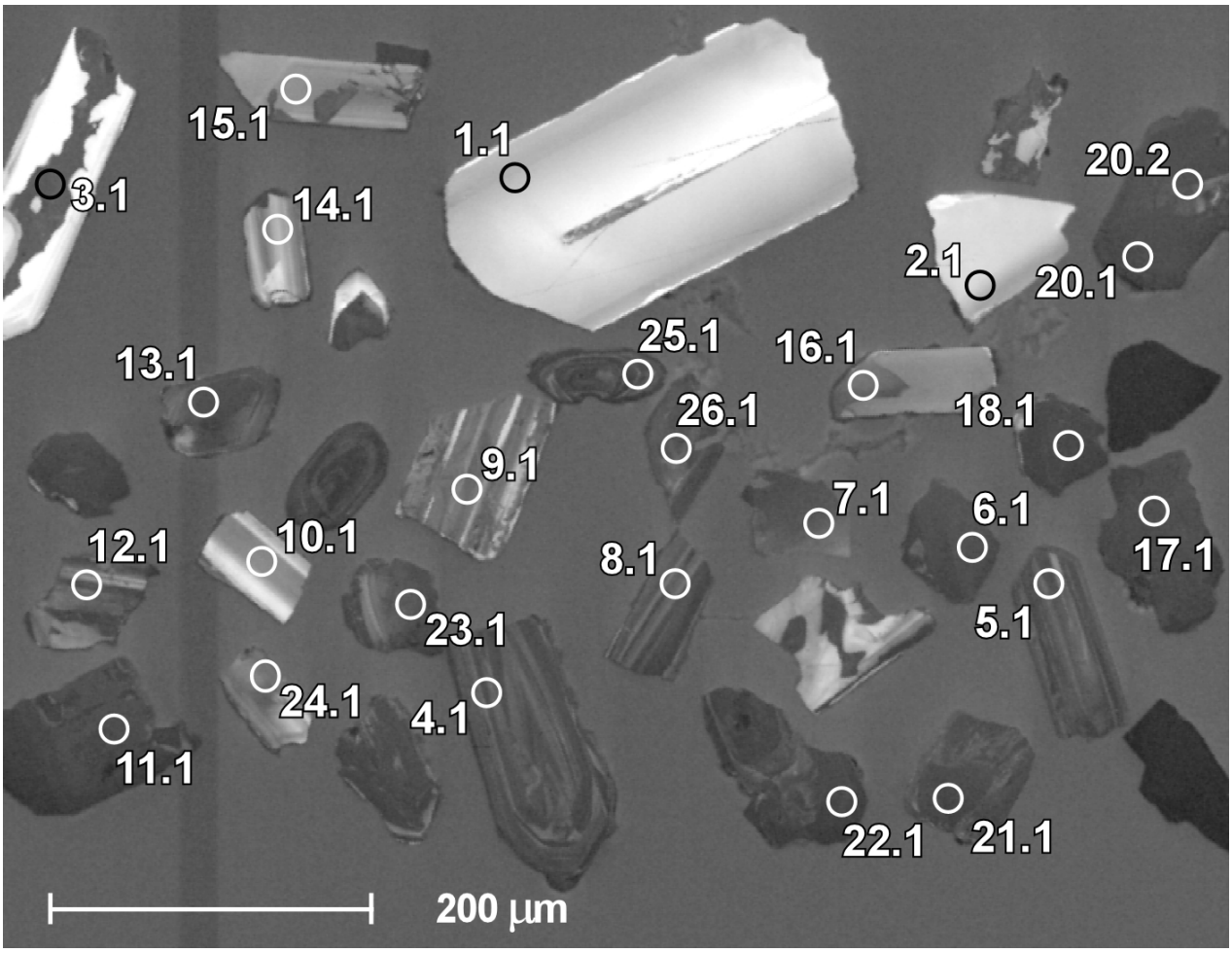

Figure 2. Cathodoluminescence image of representative zircons from sample 229285: amphibole–biotite schist, Mount Rupert. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 22–23 October 2021, using SHRIMP-B. Eight analyses of the M257 standard were obtained during the session, of which seven analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.27% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-seven analyses were obtained from 26 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3) and a probability density plot (Fig. 4).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable and considered not to be geologically significant. The remaining 26 analyses can be divided into three groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^{*}/^{206}Pb$ date of 2998 ± 18 Ma (1σ).

Group S comprises 23 analyses of 23 zircons (Table 1), which yield $^{207}Pb^{*}/^{206}Pb$ dates of 3295–2999 Ma.

Group Z comprises two analyses of two zircons (Table 1), which yield $^{207}Pb^{*}/^{206}Pb$ dates of 2635–2587 Ma.

It is possible that all of the analyses in Groups Y and S are of unmodified detrital zircons, in which case the date of 2998 ± 18 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sedimentary protolith. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^{*}/^{206}Pb$ date of 3006 ± 6 Ma (MSWD = 1.2) for the youngest eight analyses in Groups Y and S.

The data for combined Groups Y and S indicate significant age components at c. 3035 and 3005 Ma, based on contributions from approximately 15 and eight analyses, respectively (Fig. 4). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

The dates of 2635–2587 Ma for the two analyses in Group Z are interpreted to indicate processing contaminants because they are similar in age and Th/U ratios to those from the other sample (GSWA 248284, Fielding et al., 2022) in the same mount.

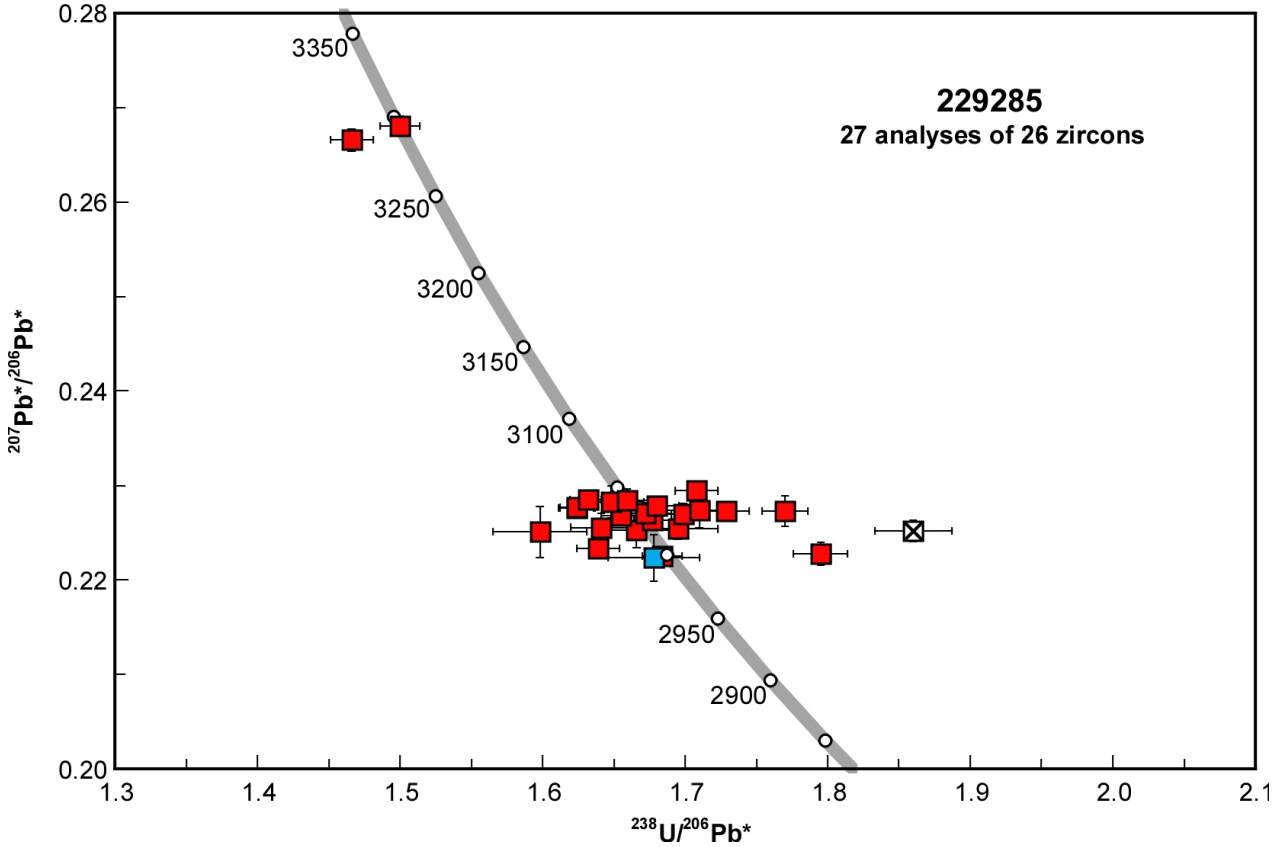

Figure 3. U–Pb analytical data for zircons from sample 229285: actinolite–biotite schist, Mount Rupert. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed square indicates Group D (discordance >5%). Two analyses in Group Z are not shown

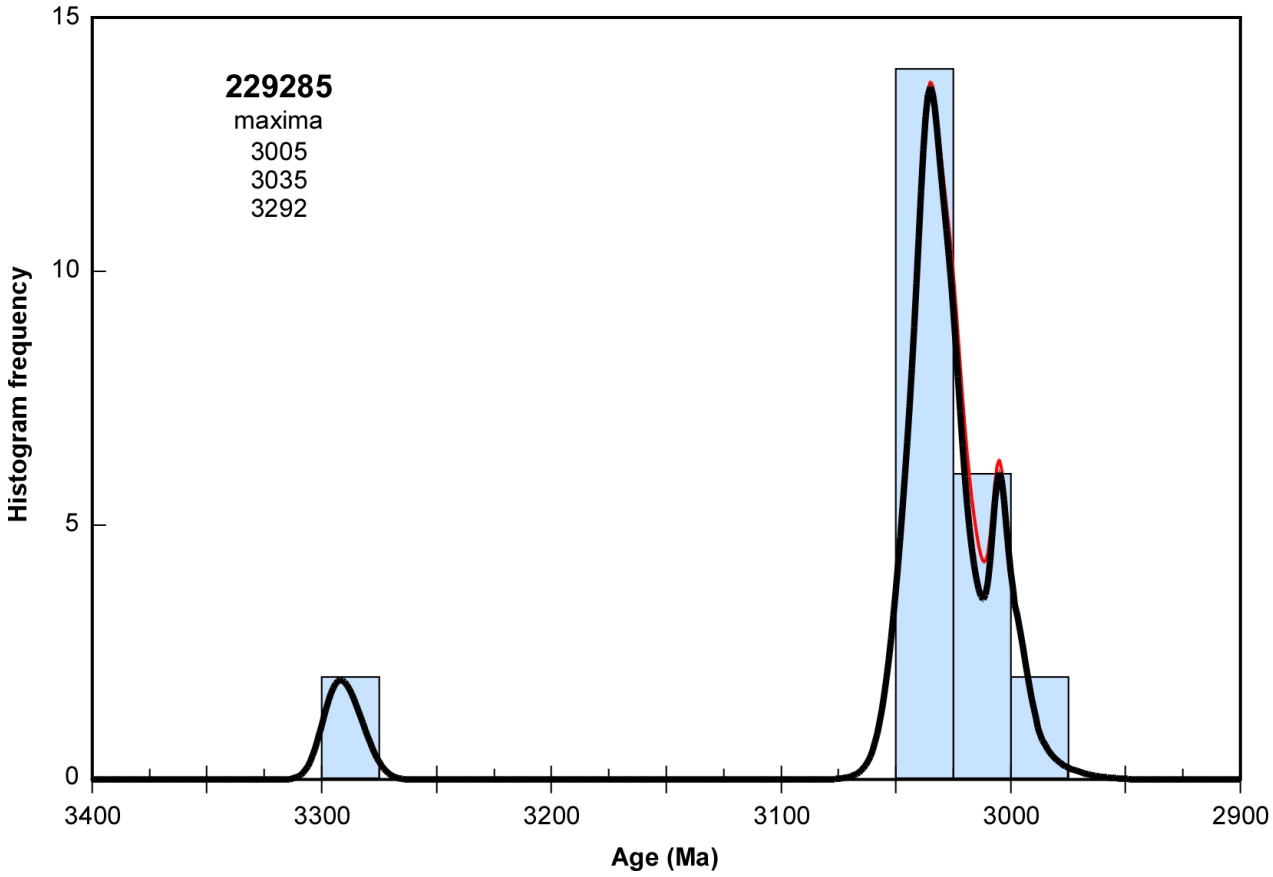

Figure 4. **Probability density diagram and histogram for sample 229285: actinolite–biotite schist, Mount Rupert. Thick curve, maxima values, and blue histogram (bin width 25 Ma) include only data with discordance <5% (24 analyses of 24 zircons). Thin curve includes all data except for those contaminants in Group Z (25 analyses of 24 zircons)**

Links

Table 1. Ion microprobe analytical results for zircons from sample 229285: actinolite–biotite schist, Mount Rupert: 229285.1.Table1.xls
SHRIMP spot images: 229285_spots.pdf
Introduction to geochronology information, 2023: Intro_2023.pdf

References

Fielding, IOH, Wingate, MTD, Ivanic, TJ and Fisher, CM 2022, 248284.1: metamonzogranite, Merino prospect; Geochronology Record 1879: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Ivanic, TJ, Wingate, MTD, Lowrey, JR and Lu, Y 2022, Formation of the Yilgarn protocraton by rift-related magmatism from 3.01 to 2.92 Ga: Geological Survey of Western Australia, Report 232, 34p.

Pidgeon, RT, Wilde, SA, Compston, W and Shield, MW 1990, Archaean evolution of the Wongan Hills greenstone belt, Yilgarn Craton, Western Australia: Australian Journal of Earth Sciences, v. 37, p. 279–292.

Quentin de Gromard, R, Ivanic, TJ and Zibra, I 2021, Pre-Mesozoic interpreted bedrock geology of the southwest Yilgarn, 2021, in Accelerated Geoscience Program extended abstracts: Geological Survey of Western Australia, Record 2021/4, p. 122–144.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Wingate, MTD, Outhwaite, MD and Lu, Y 2018, 205933: metamonzogranite dyke, Ninan prospect; Geochronology Record 1458: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

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Acknowledgment of Country

The Government of Western Australia acknowledges Aboriginal people as the traditional custodians throughout Western Australia and their enduring connection to the land, waterways and community. We pay our respects to Elders past and present.

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